

AVIRAL TRIPATHI

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SUMMARY

Full-stack MERN/PERN developer with experience in building scalable web applications, RESTful APIs, and AI-integrated cloud-deployed solutions. Strong foundation in data structures and algorithms, with proven problem-solving skills through hackathons and competitive programming.

EDUCATION

KLE Technological University, Hubballi, Karnataka

Aug 2024 – Present

Bachelor of Engineering in Computer Science CGPA: 8.34

Relevant Coursework: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECT EXPERIENCE

- **ResQRoute – Emergency Coordination Platform** [Live](#) / [GitHub](#) May 2026
Next.js 15, TypeScript, PostgreSQL, Prisma, Tailwind CSS
Built a real-time hotel emergency management system coordinating guests, staff, and first responders during fire, medical, and structural incidents.
 - Implemented a Dijkstra-driven evacuation pathfinder over a Prisma graph, applying dynamic distance and floor-penalty weighting for real-time hazard-aware rerouting.
 - Developed a multilingual speech-to-triage pipeline combining AssemblyAI transcription with OpenRouter LLM classification, generating structured JSON outputs for emergency dispatch.
 - Built an automated emergency dispatch system using Twilio SMS/Voice with room-level location tracking, plus role-based Next.js dashboards backed by Prisma + Supabase Postgres.
- **MediSumm - AI Medical Report Simplifier** [Live](#) / [GitHub](#) Apr 2026
Next.js 15, TypeScript, PostgreSQL, Prisma, pgvector, Tailwind CSS, OpenAI Embeddings
Built an AI web app that converts medical reports (labs, discharge summaries, prescriptions) into patient-friendly summaries with multilingual translation.
 - Implemented PDF + OCR pipeline using pdf-parse, Sharp, and Tesseract.js to extract structured text from reports.
 - Built a deterministic abnormal lab detector (40+ tests, gender-aware) and combined it with RAG (OpenAI embeddings + pgvector retrieval) to generate grounded summaries with citations.
- **Streamly – YouTube Clone** [GitHub](#) Jan 2026
Node.js, Express.js, MongoDB, Mongoose, Cloudinary
 - Developed a scalable video streaming backend with **RESTful APIs** for authentication, video uploads, and metadata management.
 - Implemented secure **JWT-based authentication** (access + refresh tokens) with bcrypt hashing and cookie-based session handling.
 - Engineered media upload pipelines with **Multer and Cloudinary**, and optimized video discovery using MongoDB aggregation and pagination.

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python, SQL

Frameworks/Libraries: React.js, Next.js, Express.js, Node.js, Tailwind CSS, HTML, CSS

Databases: MongoDB, MySQL, PostgreSQL

Tools/Concepts: Git, GitHub, Postman, VS Code, Vercel

ACHIEVEMENTS

- **Winner – Hack Karnataka (HealthTech Track)** – 1st place among 50+ finalist teams at a national-level hackathon.
- **Finalist – AlgoBlitz:** Achieved a Top 3 ranking in a university-wide Data Structures and Algorithms Competition.
- **Solved 500+ problems on Codeforces and LeetCode** with regular contest participation.